Fig. 1 | Three-layer audit

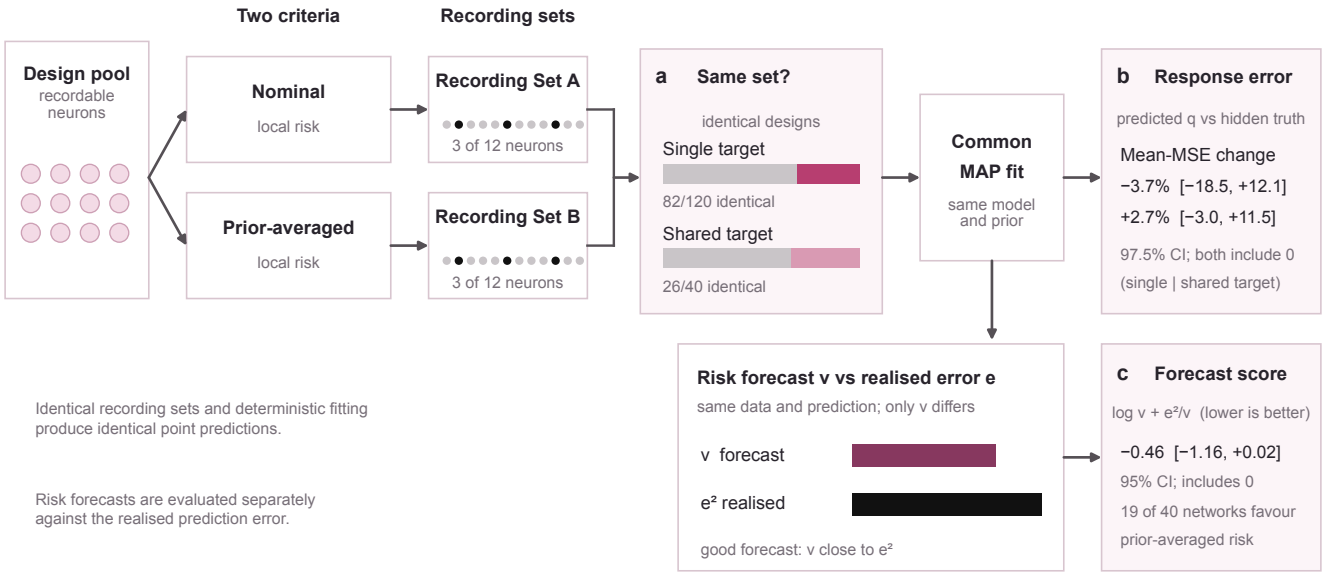


Fig. 2 | Connectivity, recordings and response



Fig. 3 | Recording decisions and response error

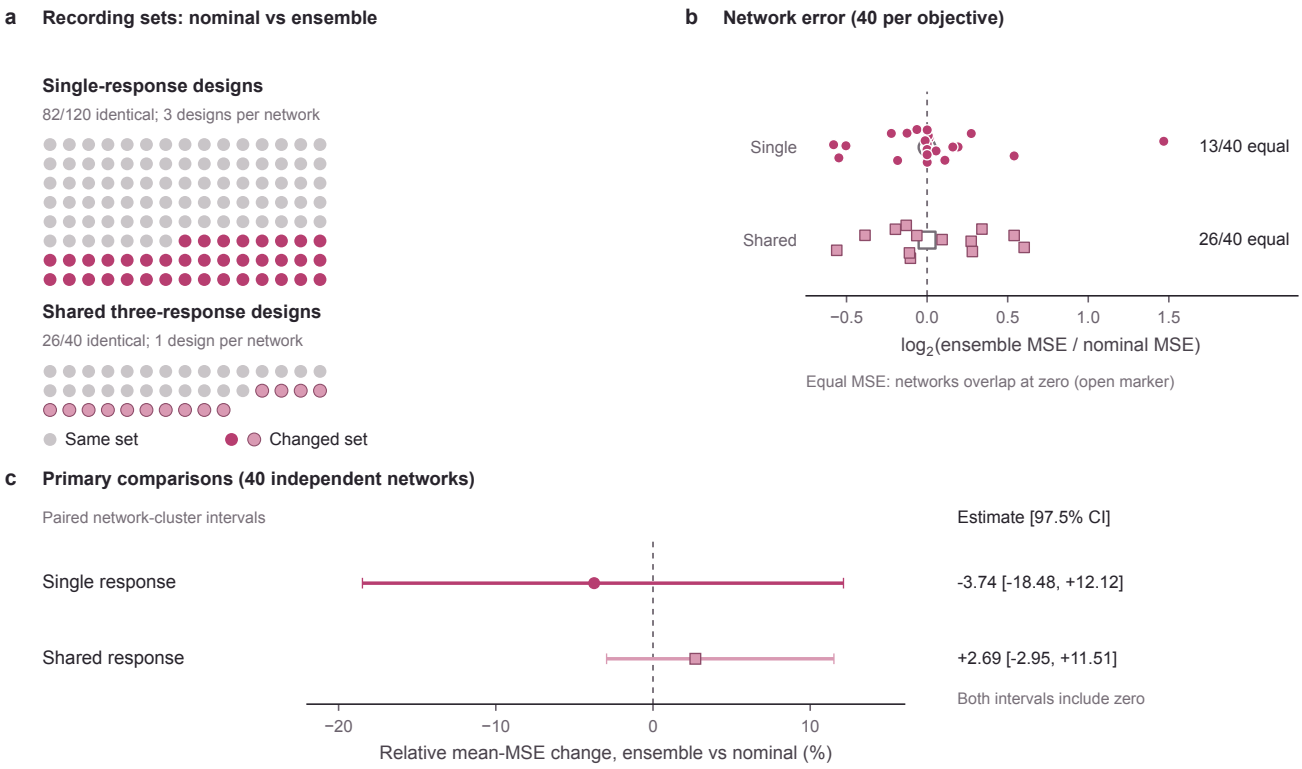
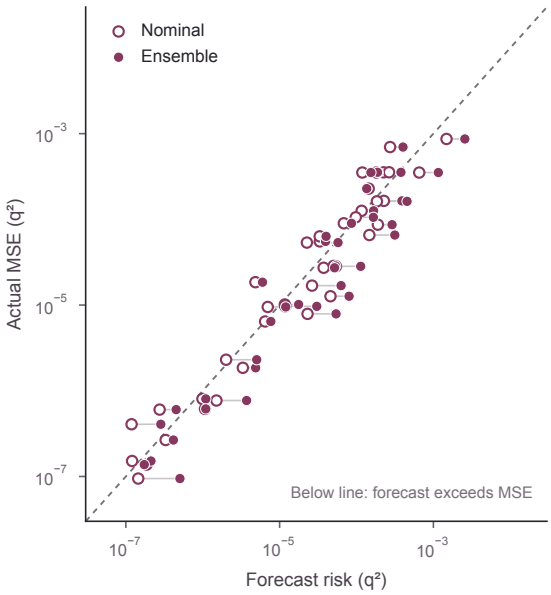
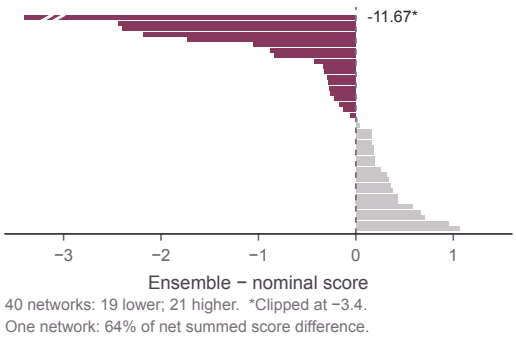


Fig. 4 | Risk forecasts on fixed predictions

a **Forecast risk and actual MSE**



b **Network-level score differences**



40 networks: 19 lower; 21 higher. *Clipped at −3.4.
One network: 64% of net summed score difference.

c **Mean score difference**

-0.46 [-1.16, +0.02], 95% network-cluster CI

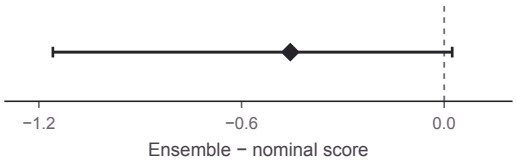
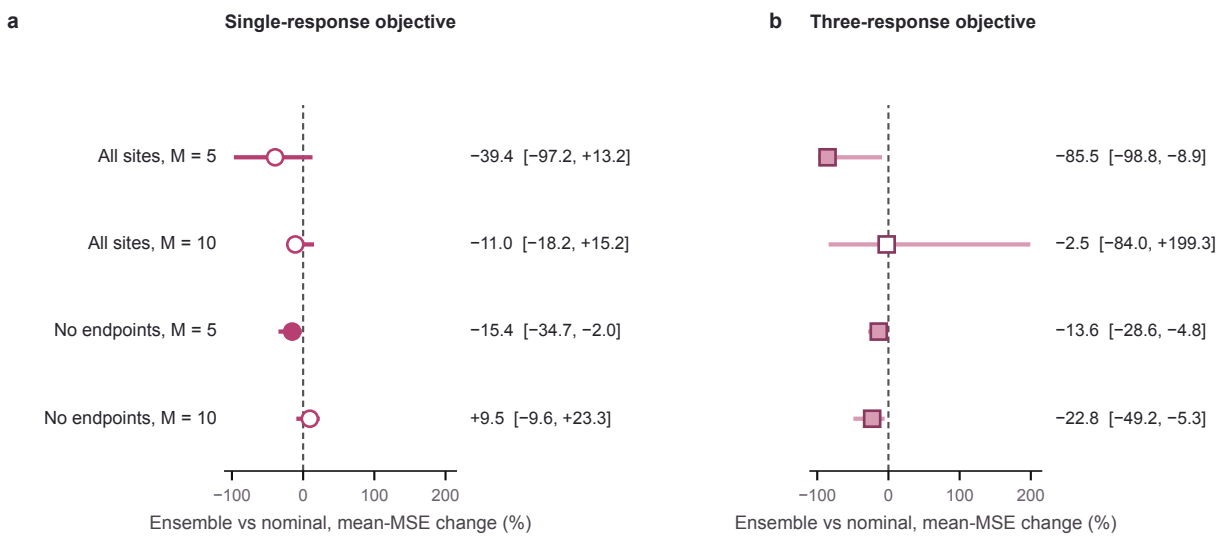


Fig. 5 | Exploratory model-family extension



Filled marker: 95% interval excludes zero (4 of 8 settings, all favouring the ensemble). Hollow: includes zero.
Exploratory: 8 independent 100-unit networks, pointwise intervals, no multiplicity adjustment; model family and N change together.

Fig. S1 | Paired distributions

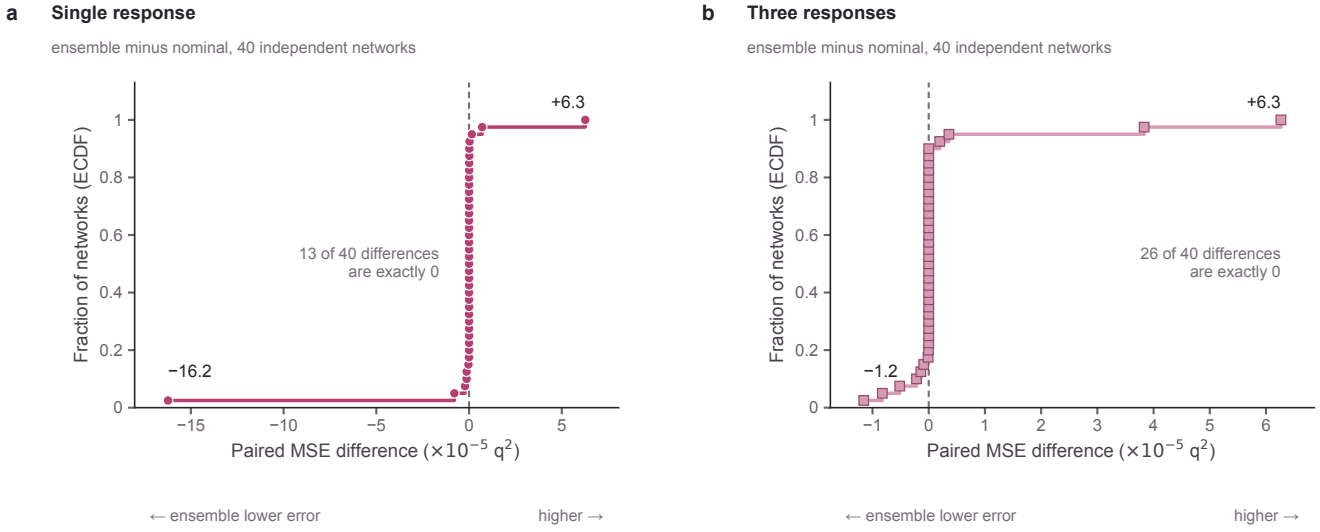
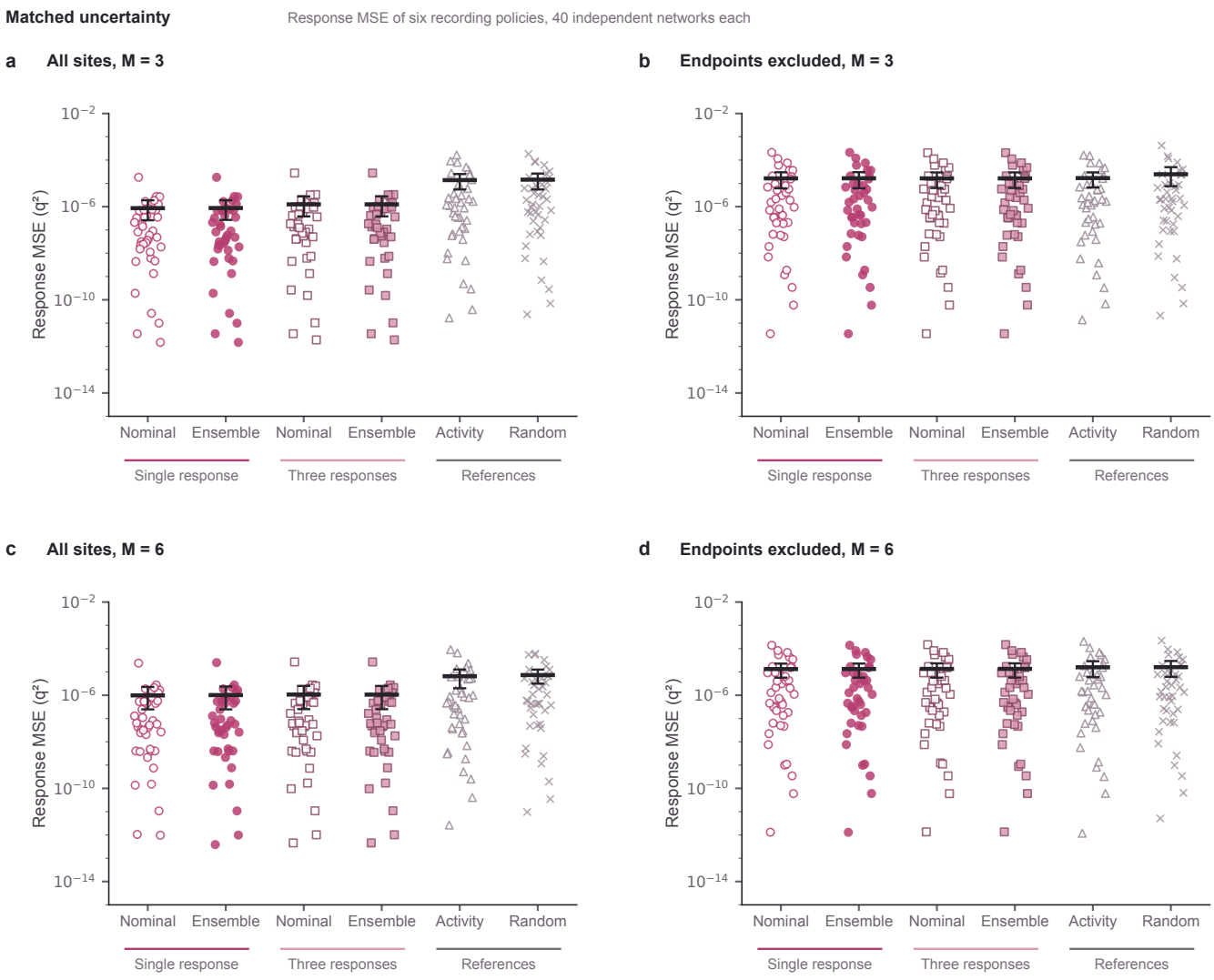
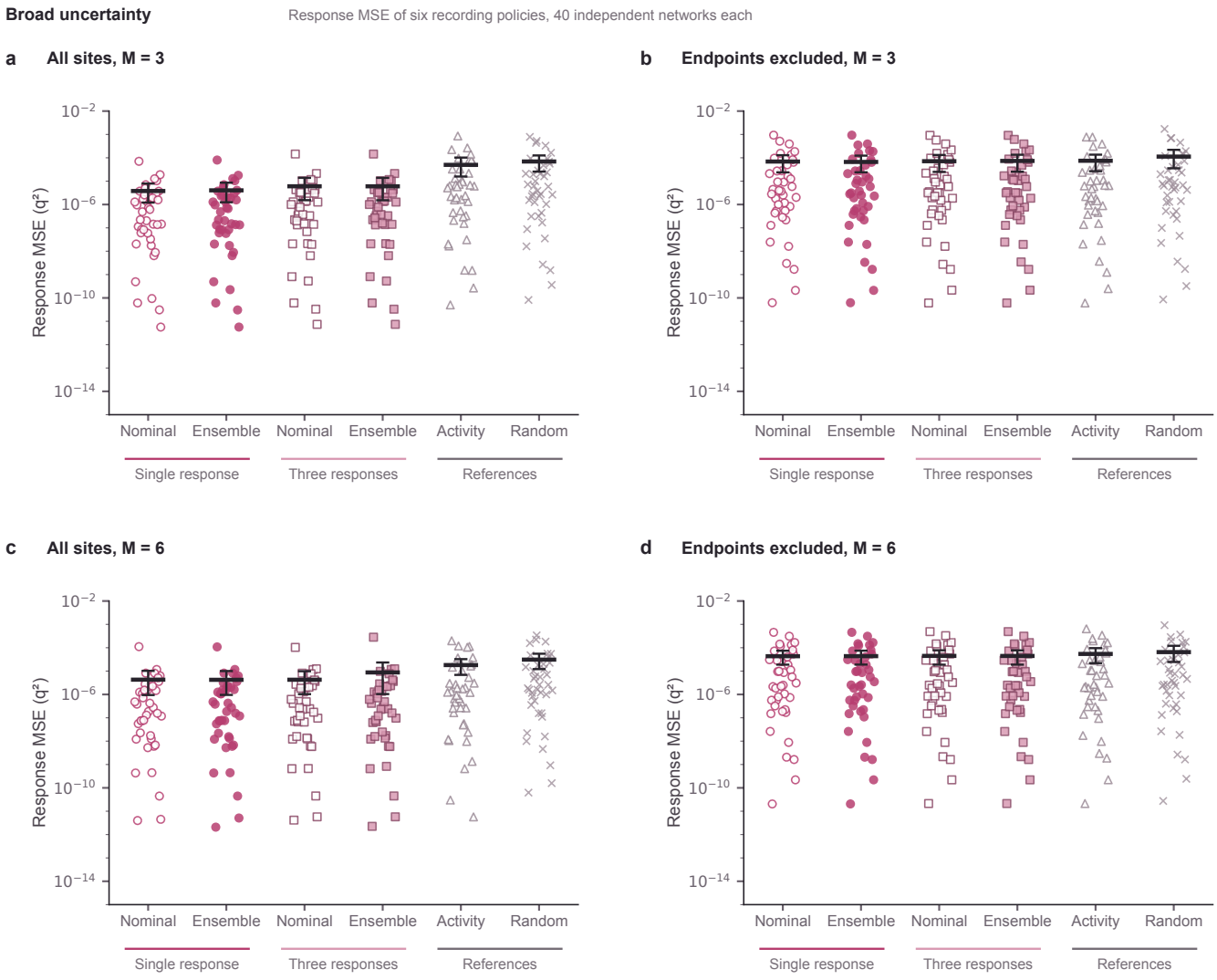


Fig. S2 | Matched uncertainty



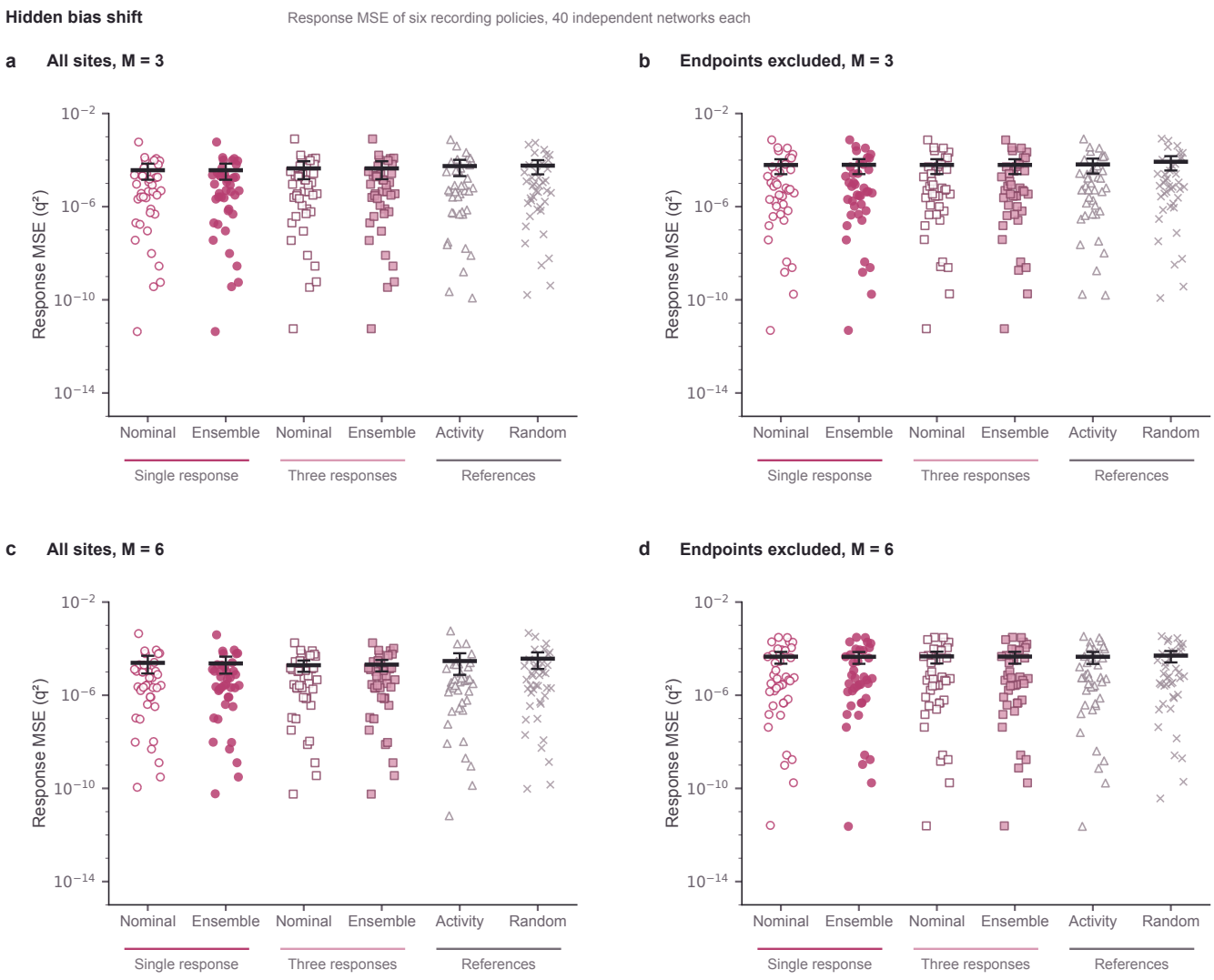
Each point is one network (truth, noise and readout repeats averaged within network). Horizontal jitter only; no point is removed.

Fig. S3 | Broad uncertainty



Each point is one network (truth, noise and readout repeats averaged within network). Horizontal jitter only; no point is removed.

Fig. S4 | Bias-shift uncertainty

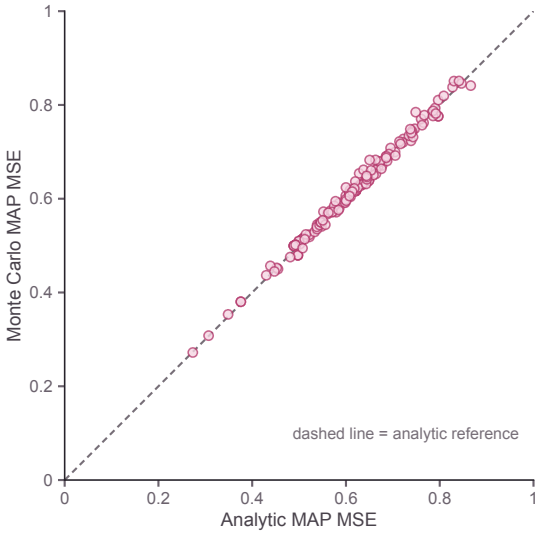


Each point is one network (truth, noise and readout repeats averaged within network). Horizontal jitter only; no point is removed.

Fig. S5 | Linear positive control

a **Analytic versus Monte Carlo MAP MSE**

Linear-Gaussian positive control, 12 independent systems



b **What the control tests**

144	target/design evaluations
12	independent linear systems
8,192	Monte Carlo draws per system

Points are not 144 independent neural networks; they are evaluations within 12 systems. No confidence intervals are drawn.

Agreement validates this computational special case, not biological assumptions.